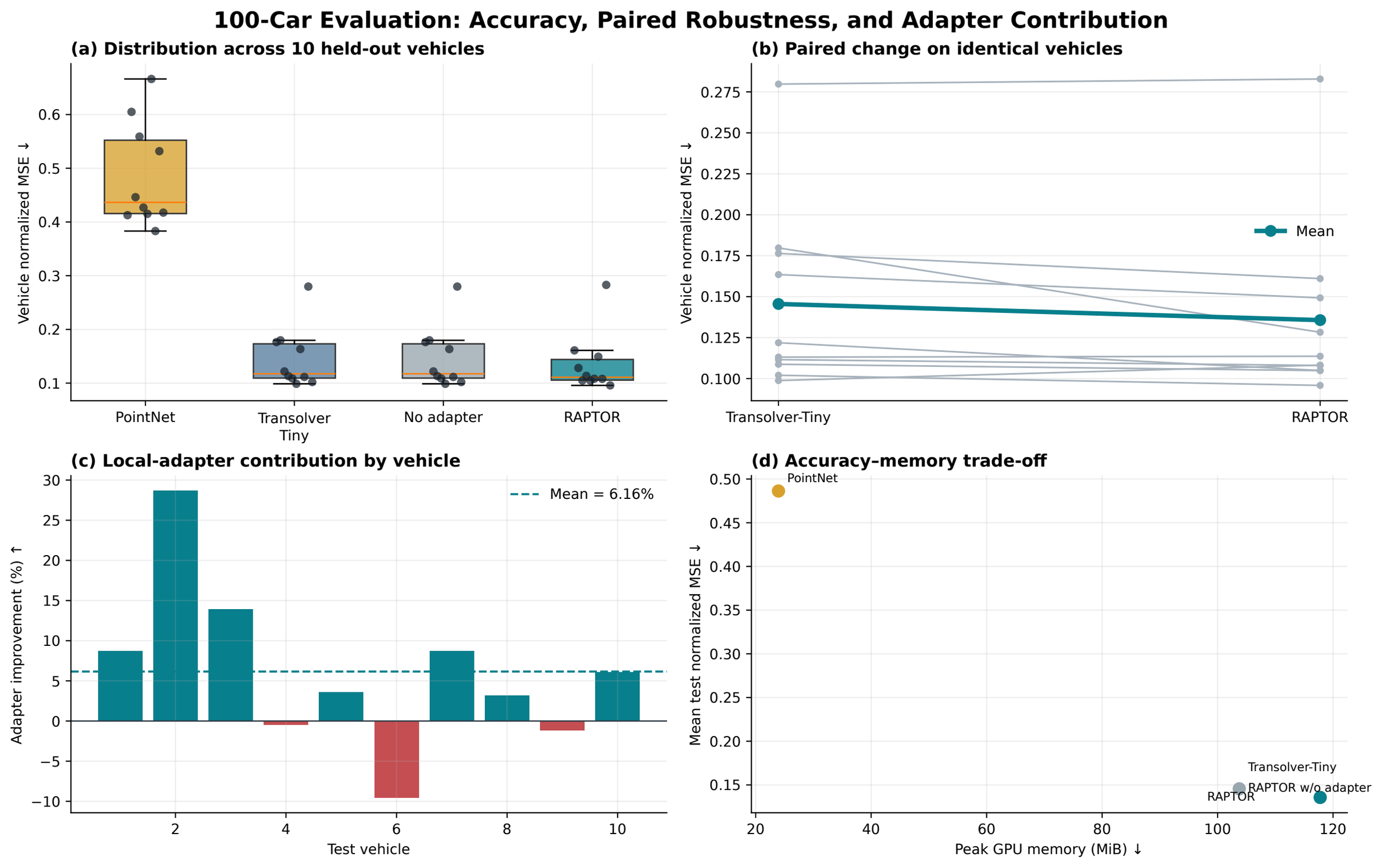

100-Car Evaluation: Accuracy, Paired Robustness, and Adapter Contribution
(a) Distribution across 10 held-out vehicles
Vehicle normalized MSE ↓
0.6
0.5
0.4
0.3
0.2
0.1
PointNet
Transolver Tiny
No adapter
RAPTOR
(b) Paired change on identical vehicles
Vehicle normalized MSE ↓
0.275
0.250
0.225
0.200
0.175
0.150
0.125
0.100
Mean
Transolver-Tiny
RAPTOR
(c) Local-adapter contribution by vehicle
Adapter improvement (%) ↑
30
25
20
15
10
5
0
−5
−10
Mean = 6.16%
2
4
6
8
10
Test vehicle
(d) Accuracy–memory trade-off
Mean test normalized MSE ↓
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
PointNet
Transolver-Tiny
RAPTOR w/o adapter
RAPTOR
20
40
60
80
100
120
Peak GPU memory (MiB) ↓